

GALAXY CLUSTERS

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GALAXIES ARE NATURALLY GATHERED.

Pulled together by their enormous gravity, they cluster tightly, sometimes orbiting one another, often colliding. As galaxies slowly move within a cluster, the cluster's structure changes. The evolution of clusters can tell astronomers about

dark matter, and clusters can even be used as cosmic “lenses” to peer back into the early universe.

TYPES OF CLUSTERS

Some galaxy clusters are sparse, loose collections of galaxies. The smallest clusters are usually termed “groups.” The Local Group (see p.328), of which the Milky Way is a member, is one such cluster. Other clusters, such as the nearby Virgo Cluster (see p.329), are denser, containing many hundreds of galaxies in a chaotic distribution. Yet other clusters, such as the Coma Cluster (see p.332), are even more dense, with galaxies settled into a neat, spherical pattern around a center dominated by giant elliptical galaxies. Although clusters differ in density, the volume of space they occupy is generally the same—a few million light-years across.

Not all galaxies exist in clusters—there are more isolated “field galaxies” than there are cluster galaxies. Some galaxy types do not exist outside clusters, however. Giant ellipticals (see p.304) always lie near the center of large clusters, as do vast, diffuse cD galaxies (below right). The most numerous cluster components may be invisible, including faint, diffuse dwarf elliptical galaxies and proposed “dark galaxies.” A dark galaxy would consist of hydrogen gas and material too thin to condense and ignite stars. The

first such galaxy may have been found in the Virgo Cluster in early 2005.

SPARSE CLUSTER

This sparse cluster, or group, of galaxies is in fact the Local Group, containing the Milky Way and its galactic neighbors. Most of the galaxies are orbiting either the Milky Way or the Andromeda Galaxy (M31).

Andromeda Galaxy (M31)

Milky Way

dense core of cluster containing many large galaxies

IDEAL DENSE CLUSTER

A dense cluster occupies the same volume as a sparse cluster such as the Local Group, but the galaxies are mainly elliptical and have a roughly spherical distribution around the cluster's center.

DENSE CLUSTER

This Hubble image shows a mix of yellow elliptical and blue-white spiral galaxies in cluster RXC J0031.1+1808, some 4 billion light-years from Earth. Fainter arcs behind this foreground cluster are created by gravitational lensing of light from far more distant galaxies.



DWARF ELLIPTICAL

Most galaxies in the Local Group, including the Sculptor Dwarf, are dwarf ellipticals. They are invisible in distant clusters but must be present.



cD GALAXY

cD galaxies are similar to giant ellipticals but have extensive, sparse outer haloes of stars. They sometimes have hints of multiple cores, suggesting the merger of several smaller ellipticals. NGC 4889 (left) is a cD galaxy at the heart of the dense Coma Cluster.